

DAX First Principles

The PakTech Mastery Guide

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1 PART 1 — DAX Foundations

1.1 What is DAX?

DAX (Data Analysis Expressions) is the formula language of Power BI. Unlike standard programming languages that manipulate logic, DAX is designed specifically to query and aggregate **Tables** and **Columns**.

1.2 Excel Formulas vs. DAX

- **Excel thinks in Cells:** $A1 + B1 = C1$. You drag formulas down.
- **DAX thinks in Columns:** $\text{Table}[\text{Sales}] + \text{Table}[\text{Cost}]$. The formula automatically applies to the entire column. There is no “dragging” in DAX.

1.3 Calculated Column vs. Measure (The Golden Rule)

This is the most important concept in Power BI. Choose wrong, and your dashboard will be slow, wrong, or broken.

Feature	Calculated Column	Measure
When does it calculate?	During data refresh.	On-the-fly when you click a chart.
Where does it live?	Physically added to your table.	Stored in memory as an equation.
Increases file size?	Yes.	No.
Best used for?	Slicers, categories, row-by-row math.	Totals, averages, percentages, KPIs.
Core Concept	Uses Row Context .	Uses Filter Context .

1.4 Row Context vs. Filter Context (Visual Explanation)

Row Context (The Assembly Line)

Imagine you are a factory worker. A box comes down the belt (Row 1). You take the Quantity inside the box, multiply it by the Price on the box, and write the total on the box. Then you move to the *next box* (Row 2). You have no idea what the total of all boxes is; you only know the math for the specific box in front of you.

Filter Context (The Tinted Window)

Imagine looking at a massive spreadsheet through a window. Suddenly, someone pulls down a blue tint over the window that says "Region = North". Instantly, all the South, East, and West rows become invisible. Now, you ask the system to `SUM(Sales)`. It only adds up the rows it can currently see.

- **Slicers** create Filter Context.
- **Visuals** (like a bar chart axis) create Filter Context.
- **DAX functions** like `CALCULATE` create Filter Context.

1.5 The Power of CALCULATE & DIVIDE

CALCULATE: Think of `CALCULATE` as a bouncer at a club. You tell the bouncer: "Go get me the sum of sales, but *only* let the 'North' region rows into the building." It is the only function that modifies filter context.

DIVIDE: Never use the `/` symbol in DAX. If the denominator is zero or blank, `/` throws an Infinity error. `DIVIDE(Numerator, Denominator, AlternateResult)` handles zero safely.

2 PART 2 — Row-Level Calculations

We use Calculated Columns here because we need the math to happen on every single receipt (row) before we can sum anything up.

2.1 A. Gross Sales

1. Business Understanding: How much money did the customer hand us at the register before any discounts were applied?

2. Mathematical Logic: The number of items they bought multiplied by the sticker price of the item.

3. DAX Concept Used: Row Context. It goes row by row.

4. Step-by-step Thinking Process:

- I need this to calculate on every single row.
- I need to look at the Quantity column and multiply it by the UnitPrice column.

5. Final DAX Formula:

```
Gross Sales = PakTech_Sales_Data[Quantity] * PakTech_Sales_Data[UnitPrice]
```

6. Why this works: Because it's a Calculated Column, DAX automatically iterates through Row 1, Row 2, Row 3, multiplying the specific quantity and price for that exact transaction.

Common Mistake

Trying to use `SUM(Quantity) * SUM(UnitPrice)`. That sums the *entire column* first, then multiplies them, resulting in a massive, mathematically incorrect number.

2.2 B. Discount Amount

1. Business Understanding: How many actual dollars are we taking off the price tag?

2. Mathematical Logic: Gross Sales multiplied by the discount percentage.

3. DAX Concept Used: Row Context.

4. Step-by-step Thinking Process:

- The dataset gives DiscountPercent as a whole number (e.g., 5, 10).
- Mathematically, 5% is 0.05. I must divide the percentage by 100 first.

- Multiply that decimal by the Gross Sales we just created.

```
Discount Amount = PakTech_Sales_Data[Gross Sales] * (PakTech_Sales_Data[DiscountPercent] / 100)
```

Common Mistake

Forgetting to divide by 100. If Gross Sales is \$1,000 and discount is 5%, doing $1000 * 5$ results in a \$5,000 discount!

2.3 C. Net Sales

1. Business Logic: The actual cash that entered our cash register. (Gross Sales minus Discount Amount).

2. DAX Concept: Row Context.

```
Net Sales = PakTech_Sales_Data[Gross Sales] - PakTech_Sales_Data[Discount Amount]
```

2.4 D. Total Cost

1. Business Logic: How much did we pay the manufacturer for the items we just sold? (Quantity multiplied by UnitCost).

2. DAX Concept: Row Context.

```
Total Cost = PakTech_Sales_Data[Quantity] * PakTech_Sales_Data[UnitCost]
```

2.5 E. Profit

1. Business Logic: Did we make money or lose money? (Net Sales minus Total Cost).

2. DAX Concept: Row Context.

```
Profit = PakTech_Sales_Data[Net Sales] - PakTech_Sales_Data[Total Cost]
```

2.6 F. Profit Margin (Row Level)

1. Business Logic: For every dollar of revenue we made, what percentage was pure profit? (Profit divided by Net Sales).

2. DAX Concept Used: Row Context + Safe Division.

3. Step-by-step Thinking Process:

- I need to divide Profit by Net Sales. What if Net Sales is \$0? The formula will break.
- I must use DIVIDE to be safe. Multiply by 100 for a percentage.

```
Profit Margin (Row) = DIVIDE(PakTech_Sales_Data[Profit],  
    PakTech_Sales_Data[Net Sales], 0) * 100
```

Why this works

DIVIDE safely handles any \$0 transactions, outputting 0 instead of throwing an error. Using the / operator would break your entire visual if a zero occurs.

3 PART 3 — Aggregated Measures

Now we switch from Calculated Columns to Measures. We are looking at the whole table at once, affected by Filter Context (clicks on the dashboard).

3.1 A. Total Net Sales & Total Profit

- 1. Business Logic:** The grand total of all revenue and profit across the whole company.
- 2. DAX Concept Used:** Filter Context + Standard Aggregation (SUM).
- 3. Thinking Process:** I don't need row-by-row math anymore. I need a grand total. I will wrap the column in a SUM function.

```
Total Net Sales = SUM(PakTech_Sales_Data[Net Sales])
```

```
Total Profit = SUM(PakTech_Sales_Data[Profit])
```

Why this works

When placed in a card visual, it sums the whole table. If you click "North" on a slicer, the Filter Context hides the other regions, and this measure instantly recalculates to only sum the North rows.

3.2 B. Total Quantity & Total Orders

- 1. Business Logic:** How many physical items left the warehouse? How many separate receipts were printed?
- 2. DAX Concept Used:** SUM and COUNTROWS.

```
Total Quantity = SUM(PakTech_Sales_Data[Quantity])
```

```
Total Orders = COUNTROWS(PakTech_Sales_Data)
```

Note: COUNTROWS returns the number of visible rows in the table. Never use SUM(OrderID).

3.3 C. Unique Customers

- 1. Business Logic:** How many *different* people shopped with us? (If Ali bought 5 times, he counts as 1 person).
- 2. DAX Concept Used:** DISTINCTCOUNT.

```
Unique Customers = DISTINCTCOUNT(PakTech_Sales_Data[CustomerID])
```

3.4 D. Average Order Value (AOV)

1. Business Logic: On average, how much money does a customer spend when they checkout? (Total Revenue divided by Total Orders).

2. DAX Concept Used: Measure Branching.

```
Average Order Value = DIVIDE([Total Net Sales], [Total Orders], 0)
```

Note: Do not use AVERAGE(Net Sales), as it averages individual rows, not distinct orders.

4 PART 4 — Logical Business Measures

4.1 A. Profit Status Classification

1. **Business Understanding:** We need to quickly label rows as High, Medium, or Low profit.
2. **Mathematical Logic:** If > 500 (High), between 200 and 500 (Medium), else (Low).
3. **DAX Concept Used:** SWITCH(TRUE()) pattern inside a **Calculated Column**.

```
Profit Status =  
SWITCH(  
    TRUE(),  
    PakTech_Sales_Data[Profit] > 500, "High",  
    PakTech_Sales_Data[Profit] >= 200, "Medium",  
    "Low"  
)
```

Exam Tip

SWITCH(TRUE()) acts like a waterfall. It checks the first condition. If true, it stops. Because > 500 was already caught by the first rule, anything falling to the second rule is inherently ≤ 500 !

4.2 B. Sales Performance Rating

1. **Business Understanding:** Is our aggregated revenue hitting the \$50,000 target?
2. **DAX Concept Used:** IF statement inside a **Measure**.

```
Sales Performance Rating =  
IF([Total Net Sales] > 50000, "Excellent", "Normal/Poor")
```

4.3 C. Region Code Mapping

1. **Business Understanding:** Map regions to numeric codes (North=1, South=2, East=3, West=4).
2. **DAX Concept Used:** Standard SWITCH formatting.

```
Region Code =  
SWITCH(  
    PakTech_Sales_Data[Region],  
    "North", 1,
```

```
"South", 2,  
"East", 3,  
"West", 4,  
0  
)
```

5 PART 5 — Advanced Filtering

CALCULATE changes the Filter Context. It overrides whatever filters are currently active on the dashboard.

5.1 A. Specific Categorical Filters

1. **Business Logic:** KPIs that *only* show North Sales, regardless of what the user clicks.
2. **DAX Concept Used:** CALCULATE(Expression, Filter)

```
North Region Sales = CALCULATE([Total Net Sales], PakTech_Sales_Data[Region] = "North")
```

Why this works

CALCULATE steps in, pulls down its own tinted window (Filter Context) to only look at the North region, and then runs the math.

5.2 B. Advanced Mathematical Filtering

1. **Business Logic:** Sum sales *only* for highly-discounted transactions (> 10%).
2. **DAX Concept Used:** CALCULATE combined with FILTER.

```
High Discount Sales =  
CALCULATE(  
    [Total Net Sales],  
    FILTER(PakTech_Sales_Data, PakTech_Sales_Data[DiscountPercent] > 10)  
)
```

Exam Tip: For complex math filters (>, <), it is best practice to explicitly use the FILTER function to iterate through the table.

6 PART 6 — ALL Concept & Percentage

1. **Business Logic:** If the North region made \$20,000, what percentage of the *entire company's* sales does that represent?

2. DAX Concept Used: CALCULATE + ALL. ALL () removes the “tinted window” entirely, ignoring visual filters.

```
Region % of Total =  
VAR Numerator = [Total Net Sales]  
VAR Denominator = CALCULATE([Total Net Sales], ALL(PakTech_Sales_Data))  
RETURN  
DIVIDE(Numerator, Denominator, 0)
```

7 PART 7 — Dashboard Design Guidance

To pass the visualization portion of your lab, you must transition from a “data processor” to a “data storyteller”.

1. Visual Selection Matrix

Do not guess which chart to use. Follow these rules:

- **KPI Cards:** Absolute totals (Total Net Sales, Unique Customers).
- **Clustered Bar Chart:** Comparisons across categories (Sales by Region).
- **Line Chart:** ONLY for time-series data (Sales over Month/Year).
- **Matrix:** Detailed cross-filtering (Rows: Region, Columns: Category).
- **Donut Chart:** Parts of a Whole. *Never use if > 4 categories.*

2. The “Z-Pattern” Layout

Humans read screens like a ‘Z’ (Top Left to Bottom Right).

- **Top Banner:** Logo, Dashboard Title, Global Slicers.
- **Top Row:** 4-5 most important KPI Cards.
- **Middle Row:** High-level charts (Bar chart, Line chart).
- **Bottom Row:** Deep-dive details (Matrix or Top 5 Table).

3. Clean BI Dashboard Rules

- **Slicers:** Change Year/Month slicers to “Dropdown” or “Tile/Button” style to save space.
- **Remove Clutter:** Turn off gridlines and clashing background colors.
- **Clear Titles:** Delete default titles (“Sum of Total Net Sales”). Rename cleanly to “Revenue by Region”.
- **Format Currency:** Format measures as Currency (\$) with 0 decimal places.